

Let $x = [1, 1, 1, 1, 3, 3, 3, 3, 1, 1, 1, 2, 2, 2, 2, 1, 1, 1, 1]$ be an input to the filter described by the I/O equation:

$$y(n) = x(n) + x(n-2) - 2x(n-3)$$

(a) Determine the impulse response $h(n)$ of this filter.

(b) Compute the corresponding output signal.

- LTI form/Convolution table/Matrix form

(c) Compute the same output using the overlap-add method of block convolution by partitioning the input into length-5 blocks.

(d) Introduce appropriate internal states, draw block diagram realization and write the corresponding sample processing algorithm.

Use MATLAB to program (b), (c) and (d), write the lab report and upload it. The report should include at least the program and the results.

解答：(a) 确定单位冲激响应 $h(n)$

根据系统的差分方程 $y(n) = x(n) + x(n-2) - 2x(n-3)$ ，令输入为单位冲激信号 $x(n) = \delta(n)$ ：

- $h(0) = \delta(0) = 1$
- $h(1) = 0$
- $h(2) = \delta(0) = 1$
- $h(3) = -2\delta(0) = -2$

结论：

$h(n) = [1, 0, 1, -2]$ ，其中 $n = 0, 1, 2, 3$ 。

(b)(c)(d)实现代码如下:

MATLAB 实现代码 (b):

```
clc; clear;

% 输入与冲激响应
x = [1, 1, 1, 1, 3, 3, 3, 3, 1, 1, 1, 2, 2, 2, 2, 1, 1, 1, 1];
h = [1, 0, 1, -2];

L = length(x);
M = length(h);
N = L + M - 1;

%% 形式1: LTI form
%  $y[n] = x[n] + x[n-2] - 2x[n-3]$ 
y_lti = zeros(1, N);
xpad = [x, zeros(1, M-1)];

for n = 1:N
    term0 = xpad(n);
    term2 = 0; term3 = 0;
    if (n-2) >= 1, term2 = xpad(n-2); end
    if (n-3) >= 1, term3 = xpad(n-3); end
    y_lti(n) = term0 + term2 - 2*term3;
end

%% 形式2: Convolution table
T = zeros(N, M); %  $T(n+1, k+1) = h[k] * x[n-k]$ 

for n = 0:N-1
    for k = 0:M-1
        idx = n - k;
        if idx >= 0 && idx < L
            T(n+1, k+1) = h(k+1) * x(idx+1);
        else
            T(n+1, k+1) = 0;
        end
    end
end

y_table = sum(T, 2).'; % 卷积表每行求和得到 y[n]
```

```

% 打印卷积表（每个 n 的4项贡献 + y[n]）
fprintf('\nConvolution table: y[n] = sum_{k=0..%d} h[k]*x[n-k]\n', M-1);
fprintf('n\t h0*x[n]\t h1*x[n-1]\t h2*x[n-2]\t h3*x[n-3]\t y[n]\n');
fprintf('-----\n');
for n = 0:N-1
    fprintf('%d\t %8g\t %10g\t %10g\t %10g\t %8g\n', ...
        n, T(n+1,1), T(n+1,2), T(n+1,3), T(n+1,4), y_table(n+1));
end

%% 形式3: Matrix form
xcol = x(:);
hcol = h(:);

c = [hcol; zeros(L-1,1)]; % 第一列 (N x 1)
r = [hcol(1); zeros(L-1,1)]; % 第一行 (L x 1)
H = toeplitz(c, r); % (N x L)

y_mat = (H * xcol).'; % 行向量

%% 直接输出三个结果
disp(' ');
disp('y_lti (LTI form) ='); disp(y_lti);
disp('y_table (Conv table) ='); disp(y_table);
disp('y_mat (Matrix form) ='); disp(y_mat);

```

MATLAB 实现代码 (c):

```
L_block = 5; % 块长度
M = length(h);
x_padded = [x, zeros(1, L_block - mod(length(x), L_block))];
y_ola = zeros(1, length(x_padded) + M - 1);

for i = 1:L_block:length(x)
    x_block = x(i:min(i+L_block-1, length(x)));
    y_block = conv(x_block, h);
    % 在重叠位置进行累加
    idx = i : i + length(y_block) - 1;
    y_ola(idx) = y_ola(idx) + y_block;
end

% 截取有效部分
y_ola = y_ola(1:length(x) + M - 1);

% 显示结果
fprintf('(c) 重叠相加法输出y(n):\n');
disp(y_ola);
```

MATLAB 实现代码 (d):

```
% 初始化状态
w1 = 0; w2 = 0; w3 = 0;
x_ext = [x, 0, 0, 0]; % 补零以清空系统状态
y_sample = zeros(1, length(x_ext));

for n = 1:length(x_ext)
    curr_x = x_ext(n);
    % 计算当前输出
    y_sample(n) = curr_x + w2 - 2 * w3;
    % 更新状态
    w3 = w2;
    w2 = w1;
    w1 = curr_x;
end

% 显示结果
fprintf('(d) 逐样点处理输出y(n):\n');
disp(y_sample);
```

运行结果及分析:

(b)

Convolution table: $y[n] = \sum_{k=0..3} h[k]*x[n-k]$																					
n	$h0*x[n]$	$h1*x[n-1]$	$h2*x[n-2]$	$h3*x[n-3]$	$y[n]$																
0	1	0	0	0	1																
1	1	0	0	0	1																
2	1	0	1	0	2																
3	1	0	1	-2	0																
4	3	0	1	-2	2																
5	3	0	1	-2	2																
6	3	0	3	-2	4																
7	3	0	3	-6	0																
8	1	0	3	-6	-2																
9	1	0	3	-6	-2																
10	1	0	1	-6	-4																
11	2	0	1	-2	1																
12	2	0	1	-2	1																
13	2	0	2	-2	2																
14	2	0	2	-4	0																
15	1	0	2	-4	-1																
16	1	0	2	-4	-1																
17	1	0	1	-4	-2																
18	1	0	1	-2	0																
19	0	0	1	-2	-1																
20	0	0	1	-2	-1																
21	0	0	0	-2	-2																
y_lti (LTI form) =																					
1	1	2	0	2	2	4	0	-2	-2	-4	1	1	2	0	-1	-1	-2	0	-1	-1	-2
y_table (Conv table) =																					
1	1	2	0	2	2	4	0	-2	-2	-4	1	1	2	0	-1	-1	-2	0	-1	-1	-2
y_mat (Matrix form) =																					
1	1	2	0	2	2	4	0	-2	-2	-4	1	1	2	0	-1	-1	-2	0	-1	-1	-2

(c)(d)

(c) 重叠相加法输出 $y(n)$:																					
1	1	2	0	2	2	4	0	-2	-2	-4	1	1	2	0	-1	-1	-2	0	-1	-1	-2
(d) 逐样点处理输出 $y(n)$:																					
1	1	2	0	2	2	4	0	-2	-2	-4	1	1	2	0	-1	-1	-2	0	-1	-1	-2

所有方案结果一致

框图:

